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EXPANDABLE INTRODUCER APPARATUS AND METHODS OF GUIDING AN OBJECT

Abstract

An expandable introducer apparatus comprising an inner tubular layer comprising an inner lumen and a radially expandable region extending along an elongated axis of the inner lumen and configured to at least partially radially expand relative to the elongated axis from a contracted orientation to an expanded orientation. An outer tubular layer comprising an inner surface bonded to the inner tubular layer, wherein a distal segment of the inner tubular layer extends beyond a distal end of the outer tubular layer. A flexible tubular member comprising a bonded region and a flexible region, wherein an inner surface of the bonded region is bonded to an outer surface of the distal segment of the inner tubular layer, and the flexible region is not bonded to the radially expandable region of the inner tubular layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/559,977, filed Mar. 1, 2024, the entire content of which is incorporated herein by reference.

FIELD

[0002] The present disclosure relates generally to an expandable introducer apparatus and methods of guiding an object, and more particularly to expandable introducer apparatus configured to introduce an object into the vasculature of a patient and methods of guiding an object through an expandable introducer apparatus.

BACKGROUND

[0003] During medical procedures, for example, a transcatheter aortic valve replacement, a plurality of objects are commonly introduced into a patient's vasculature. Such objects may include transcatheter valve prosthesis delivery systems. These delivery systems may include various components, for example, a catheter assembly which includes a collapsed valve prosthesis, a guide wire and a plurality of sheaths and tubes. In order to provide percutaneous access for the objects into the patient's vasculature, an introducer is typically employed. Conventional introducers can comprise a hub component, a hemostasis valve, and an elongated tubular member which forms a lumen that runs through a central axis of the elongated tubular member through which the objects enter and exit the patient's vasculature.

[0004] During an aortic valve replacement procedure, introducers can accommodate objects of various sizes and shapes while also providing and maintaining support for the objects when they are inserted, removed and/or adjusted. Furthermore, introducers can be reliably percutaneously inserted without damaging the patient's vasculature while reducing the complexity of introducing and removing objects from the vasculature during surgical procedures.

[0005] In light of the above, a need exists for an expandable introducer with a flexible tubular member having the capability of expanding and contracting to accommodate objects during medical procedures while maintaining an atraumatic structure to minimize damage to the patient's vasculature.

SUMMARY

[0006] The following presents a simplified summary of the disclosure to provide a basic understanding of some aspects described in the detailed description.

[0007] Features of the present disclosure provide an expandable introducer apparatus having a flexible tubular member that seals to an outer surface of an object as it passes therethrough. Providing an introducer apparatus with the flexible tubular member can allow a clinician to reliably insert and remove objects from the vasculature of a patient while reducing injury to the patient's vasculature and reducing or eliminating blood loss.

[0008] In aspects, an expandable introducer apparatus comprises an inner tubular layer comprising an inner lumen and a radially expandable region extending along an elongated axis of the inner lumen and configured to at least partially radially expand relative to the elongated axis from a contracted orientation to an expanded orientation. The expandable introducer apparatus further comprises an outer tubular layer comprising an inner surface bonded to the inner tubular layer, wherein a distal segment of the inner tubular layer extends beyond a distal end of the outer tubular

layer. The expandable introducer apparatus still further comprises a flexible tubular member comprising a bonded region and a flexible region, wherein an inner surface of the bonded region is bonded to an outer surface of the distal segment of the inner tubular layer. The flexible region is not bonded to the radially expandable region of the inner tubular layer along an axial length of the distal segment. Furthermore, the radially expandable region of the inner tubular layer along the axial length of the distal segment extends under the flexible region of the flexible tubular member. [0009] In further aspects, a method of guiding an object through an inner lumen of an expandable introducer apparatus is provided. The expandable introducer apparatus comprises an inner tubular layer comprising the inner lumen and a radially expandable region extending along an elongated axis of the inner lumen. The expandable introducer apparatus further comprises an outer tubular layer comprising an inner surface bonded to the inner tubular layer, wherein a distal segment of the inner tubular layer extends beyond a distal end of the outer tubular layer. The expandable introducer apparatus further comprises a flexible tubular member comprising a bonded region and a flexible region, wherein an inner surface of the bonded region is bonded to an outer surface of the distal segment of the inner tubular layer. The flexible region is not bonded to the radially expandable region of the inner tubular layer along an axial length of the distal segment. Furthermore, the radially expandable region of the inner tubular layer along the axial length of the distal segment extends under the flexible region of the flexible tubular member. The method comprises distally advancing the object through a proximal end of the inner lumen, wherein a portion of the radially expandable region radially expands from a contracted orientation to an expanded orientation while a corresponding portion of the outer tubular layer expands to accommodate the expanded orientation of the radially expandable region. The method further comprises distally advancing the object past the distal end of the outer tubular layer to radially expand the radially expandable region of the inner tubular layer along the axial length of the distal segment while simultaneously circumferentially stretching the flexible region of the flexible tubular member, wherein the distal segment of the inner tubular layer and the flexible tubular member simultaneously expand to accommodate the object passing through the distal segment of the inner tubular layer. The method further comprises distally advancing the object past a distal end of the inner tubular layer.

[0010] Additional features and advantages of the aspects disclosed herein will be set forth in the detailed description that follows, and in part will be clear to those skilled in the art from that description or recognized by practicing the aspects described herein, including the detailed description which follows, the claims, as well as the appended drawings. It is to be understood that both the foregoing general description and the following detailed description present aspects intended to provide an overview or framework for understanding the nature and character of the aspects disclosed herein. The accompanying drawings are included to provide further understanding and are incorporated into and constitute a part of this specification. The drawings illustrate various aspects of the disclosure, and together with the description explain the principles and operations thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] These and other features, aspects and advantages are better understood when the following detailed description is read with reference to the accompanying drawings, in which:

[0012] FIG. 1 is a schematic view of an exemplary expandable introducer apparatus in accordance with aspects of the present disclosure;

[0013] FIG. 2 is a schematic enlarged view of a distal end portion of the expandable introducer apparatus taken at view 2 of FIG. 1 in accordance with aspects of the present disclosure;

[0014] FIG. 3 is top schematic view of the distal end portion of the expandable introducer apparatus along line 3-3 of FIG. 2;

[0015] FIG. 4 is a schematic cross-sectional view of the distal end portion taken at line 4-4 of FIG. 2;

[0016] FIG. 5 is a schematic cross-sectional view of a distal segment of the distal end portion taken at line 5-5 of FIG. 2;

[0017] FIG. 6 is a schematic cross-sectional view of another embodiment of a distal segment of the distal end portion taken at line 5-5 of FIG. 2;

[0018] FIG. 7 is a schematic cross-sectional view of another embodiment of a distal segment of the distal end portion taken at line 5-5 of FIG. 2;

[0019] FIG. 8 is a schematic cross-sectional view of a flexible tip taken at line 8-8 of FIG. 2;

[0020] FIGS. 9-12 illustrate example steps in methods of manufacturing the expandable introducer apparatus in accordance with the present disclosure;

[0021] FIG. 13 illustrates a schematic perspective view of an exemplary support member that can be provided with the distal segments illustrated in FIGS. 5-7;

[0022] FIG. 14 illustrates a schematic side view of the support member of FIG. 13 in a two-dimensional planar state;

[0023] FIG. 15 illustrates a schematic perspective view of another embodiment of a support member that can be provided with the distal segments illustrated in FIGS. 5-7;

[0024] FIG. 16 illustrates a schematic side view of the support member of FIG. 15 in a two-dimensional planar state;

[0025] FIG. 17 illustrates a schematic perspective view of another embodiment of a support member that can be provided with the distal segments illustrated in FIGS. 5-7;

[0026] FIG. 18 illustrates a schematic side view of the support member of FIG. 17 in a two-dimensional planar state;

[0027] FIG. 19 illustrates a schematic top view of the expandable introducer apparatus with the support member of FIG. 13 circumferentially distributed around an elongated axis of the expandable introducer apparatus;

[0028] FIG. 20 illustrated a schematic bottom view of the expandable introducer apparatus with the support member of FIG. 13 circumferentially distributed around the elongated axis of the expandable introducer apparatus;

[0029] FIG. 21 illustrates a schematic cross-sectional view generally taken at line 21-21 of FIG. 19 showing a distribution of peaks of the support member of FIG. 13 around the elongated axis of the expandable introducer in accordance with the present discloser;

[0030] FIG. 22 illustrates a schematic top view of the expandable introducer apparatus with the support member of FIG. 15 circumferentially distributed around the elongated axis of the expandable introducer apparatus;

[0031] FIG. 23 illustrates a schematic bottom of the expandable introducer apparatus with the support member of FIG. 15 circumferentially distributed around the elongated axis of the expandable introducer apparatus;

[0032] FIG. 24 illustrates a schematic cross-sectional view generally taken at line 24-24 of FIG. 22 showing a distribution of peaks of the support member of FIG. 15 around the elongated axis of the expandable introducer apparatus in accordance with the present discloser;

[0033] FIG. 25 illustrate a schematic top view of the expandable introducer apparatus with the support members of FIGS. 13, 15, and 17 circumferentially distributed around a bottom portion of the elongated axis of the expandable introducer apparatus;

[0034] FIG. 26 illustrates a schematic bottom view of the expandable introducer apparatus with the support members of FIGS. 13, 15, and 17 circumferentially distributed around a bottom portion of the elongated axis of the expandable introducer apparatus;

[0035] FIG. 27 illustrates a schematic cross-sectional view generally taken at line 27-27 of FIG. 25

showing a distribution of peaks of the support members of FIGS. 13, 15, and 17 around the bottom portion of the elongated axis of the expandable introducer apparatus in accordance with the present discloser;

[0036] FIG. 28 illustrates a schematic top view of the expandable introducer apparatus with the support members of FIGS. 13, 15, and 17 biased from radially flaring outward by a constriction of a flexible tubular member;

[0037] FIG. 29 illustrates a schematic top view of the expandable introducer apparatus of FIG. 28 with a distal end being radially flared out relative to a proximal end of the support members against a bias of the constriction of the flexible tubular member;

[0038] FIG. 30 schematically illustrates an object passing through an inner lumen of an inner tubular layer wherein a portion of a radially expandable region of the inner tubular layer radially expands;

[0039] FIG. 31 is a cross-sectional view taken at line 31-31 of FIG. 30, wherein an elongated split of the outer tubular layer is shown to separate to accommodate the expanded orientation of the radially expandable region;

[0040] FIG. 32 schematically illustrates the object passing through a distal segment of the inner tubular layer, wherein the distal segment of the inner tubular layer and a flexible tubular member simultaneously expand to accommodate the object passing through the distal segment;

[0041] FIG. 33 is a cross-sectional view taken at line 33-33 of FIG. 32;

[0042] FIG. 34 schematically illustrates the expandable introducer apparatus in accordance with the present disclosure after the object has passed through a distal end of the expandable introducer; and

[0043] FIG. 35 illustrates an exemplary flow chart demonstrating a method of utilizing the expandable introducer apparatus.

DETAILED DESCRIPTION

[0044] Aspects will now be described more fully hereinafter with reference to the accompanying drawings in which example aspects are shown. Whenever possible, the same reference numerals are used throughout the drawings to refer to the same or like parts. However, this disclosure may be embodied in many different forms and should not be construed as limited to the aspects set forth herein.

[0045] As used herein, the term “about” means that amounts, sizes, formulations, parameters, and other quantities and characteristics are not, and need not be, exact, but may be approximate and/or larger or smaller, as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art.

[0046] Ranges can be expressed herein as from “about” one value, and/or to “about” another value. When such a range is expressed, aspects include from the one value to the other value. Similarly, when values are expressed as approximations by use of the antecedent “about,” it will be understood that the value forms another aspect. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

[0047] Directional terms as used herein—for example up, down, right, left, front, back, top, bottom, upper, lower, etc.—are made only with reference to the figures as drawn and are not intended to imply absolute orientation.

[0048] Unless otherwise expressly stated, it is in no way intended that any methods set forth herein be construed as requiring that its steps be performed in a specific order, nor that with any apparatus, specific orientations be required. Accordingly, where a method claim does not actually recite an order to be followed by its steps, or that any apparatus claim does not actually recite an order or orientation to individual components, or it is not otherwise specifically stated in the claims or description that the steps are to be limited to a specific order, or that a specific order or orientation to components of an apparatus is not recited, it is in no way intended that an order or orientation be inferred in any respect. This holds for any possible non-express basis for interpretation, including

matters of logic relative to arrangement of steps, operational flow, order of components, or orientation of components; plain meaning derived from grammatical organization or punctuation, and; the number or type of aspects described in the specification.

[0049] As used herein, the singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise. Thus, for example, reference to “a” component includes aspects having two or more such components, unless the context clearly indicates otherwise.

[0050] The word “exemplary,” “example,” or various forms thereof are used herein to mean serving as an example, instance, or illustration. Any aspect or design described herein as “exemplary” or as an “example” should not be construed as preferred or advantageous over other aspects or designs. Furthermore, examples are provided solely for purposes of clarity and understanding and are not meant to limit or restrict the disclosed subject matter or relevant portions of this disclosure in any manner. It can be appreciated that a myriad of additional or alternate examples of varying scope could have been presented but have been omitted for purposes of brevity.

[0051] As used herein, the terms “comprising,” “including,” and variations thereof shall be construed as synonymous and open-ended, unless otherwise indicated. A list of elements following the transitional phrases comprising or including is a non-exclusive list, such that elements in addition to those specifically recited in the list may also be present.

[0052] The terms “substantial,” “substantially,” and variations thereof as used herein are intended to represent that a described feature is equal or approximately equal to a value or description. Moreover, “substantially” is intended to denote that two values are equal or approximately equal. The term “substantially” may denote values within about 10% of each other, for example, within about 5% of each other, or within about 2% of each other.

[0053] Modifications may be made to the instant disclosure without departing from the scope or spirit of the claimed subject matter. Unless specified otherwise, “first,” “second,” or the like are not intended to imply a temporal aspect, a spatial aspect, an ordering, etc. Rather, such terms are merely used as identifiers, names, etc. for features, elements, items, etc. For example, a first end and a second end generally correspond to end A and end B or two different ends.

[0054] Unless otherwise indicated, the terms “distal” and “proximal” are used in the following description with respect to a position or direction relative to the treating clinician. “Distal” and “distally” are positions distant from or in a direction away from the clinician, and “proximal” and “proximally” are positions near or in a direction toward the clinician. In addition, the term “self-expanding” may be used in the following description with reference to one or more valve or stent structures of the prostheses hereof and is intended to convey that the structures are shaped or formed from a material that can be provided with a mechanical memory to return the structure from a compressed or constricted delivery configuration to an expanded deployed configuration or vice versa. Non-exhaustive exemplary self-expanding materials include stainless steel, a pseudo-elastic metal such as a nickel titanium alloy or nitinol, various polymers, or a so-called super alloy, which may have a base metal of nickel, cobalt, chromium, or other metal. Mechanical memory may be imparted to a wire or stent structure by thermal treatment to achieve a spring temper in stainless steel, for example, or to set a shape memory in a susceptible metal alloy, such as nitinol. Various polymers that can be made to have shape memory characteristics may also be suitable for use in aspects hereof to include polymers such as polynorborene, trans-polyisoprene, styrene-butadiene, and polyurethane. As well poly L-D lactic copolymer, oligo caprylactone copolymer and poly cyclo-octine can be used separately or in conjunction with other shape memory polymers.

[0055] Diseases associated with heart valves, such as those caused by damage or a defect, can include stenosis and valvular insufficiency or regurgitation. For example, valvular stenosis causes the valve to become narrowed and hardened which can prevent blood flow to a downstream heart chamber from occurring at the proper flow rate and may cause the heart to work harder to pump the blood through the diseased valve. Valvular insufficiency or regurgitation occurs when the valve

does not close completely, allowing blood to flow backwards, thereby causing the heart to be less efficient. A diseased or damaged valve, which can be congenital, age-related, drug-induced, or in some instances, caused by infection, can result in an enlarged, thickened heart that loses elasticity and efficiency. Some symptoms of heart valve diseases can include weakness, shortness of breath, dizziness, fainting, palpitations, anemia and edema, and blood clots which can increase the likelihood of stroke or pulmonary embolism. Symptoms can often be severe enough to be debilitating and/or life threatening.

[0056] Heart valve prostheses have been developed for repair and replacement of diseased and/or damaged heart valves. Such heart valve prostheses can be percutaneously delivered and deployed at the site of the diseased heart valve through catheter-based delivery systems. Such heart valve prostheses generally include a frame or stent and a prosthetic valve mounted within the frame. Such heart valve prostheses are delivered in a radially compressed or crimped configuration so that the heart valve prosthesis can be advanced through the patient's vasculature. Once positioned at the treatment site, the heart valve prosthesis is expanded to engage tissue at the diseased heart valve region to, for instance, hold the heart valve prosthesis in position.

[0057] In order to facilitate the introduction of the catheter-based delivery system including the heart valve prostheses and various other objects utilized during a procedure, an expandable introducer is first inserted through the skin of the patient. The expandable introducer is typically one which has an expandable region to accommodate the passage of the objects therethrough.

[0058] FIG. 1 illustrates an expandable introducer **101** for use with a plurality of objects, such as for example, a catheter-based delivery system. The expandable introducer **101** can be delivered to a desired location within a patient's vasculature for inserting objects therethrough. As shown in FIG. 1, the expandable introducer **101** can comprise an outer tubular layer **401**, an inlet **105**, and a flush port **111**. The object can be inserted through the proximal end of the inlet **105** and then advanced distally along a length of the expandable introducer. The object then exits the expandable introducer **101** through a distal end portion of the expandable introducer **101** where the object is thereafter advanced within the vasculature to the treatment site.

[0059] A distal end portion **103** of the expandable introducer **101** is illustrated and discussed with initial reference to FIGS. 2-5. As shown in FIGS. 4-5, an inner tubular layer **403** comprises an inner lumen **405** and a radially expandable region **407** (see FIG. 4) extending along an elongated axis **201** of the inner lumen **405**. The radially expandable region **407** may be configured to at least partially radially expand relative to the elongated axis **201** from a contracted orientation (e.g., see FIGS. 4-5) to an expanded orientation (e.g., see FIGS. 31 and 33). The inner tubular layer **403** may be a wide range of materials. For example, the inner tubular layer **403** may be formed from a flexible material, such as for example, tetrafluoroethylene (TFE), Teflon®, polytetrafluoroethylene (PTFE), polyethylene, polyethylene terephthalate (PET), polyester or the like. Further, the inner tubular layer **403** may have a low coefficient of friction to aid in the insertion of objects therethrough.

[0060] In some aspects, the radially expandable region **407** of the inner tubular layer **403** can comprise a foldable region **409** that may be configured to at least partially unfold from a contracted orientation to an expanded orientation. In some aspects, the foldable region **409** further comprises an outer fold **411** and an inner fold **413** in the contracted orientation as shown in FIG. 4. The outer fold **411** and the inner fold **413** are configured such that when an object **3101** (e.g., FIG. 14) passes through the inner lumen **405**, the outer fold **411** and the inner fold **413** begin to unfold. For example, the outer fold **411** and the inner fold **413** can partially or completely unfold to allow the inner tubular layer **403** to radially expand to accommodate the object **3101** passing through the inner lumen **405**.

[0061] Throughout the disclosure, the various embodiments of the expandable introducer is illustrated and described being used to introduce the object **3101** schematically illustrated in the drawings. The schematic illustration of the object **3101** is not intended to be limited in size and

shape to the illustrated size and shape but may take the size and shape of the object being introduced. Such objects may comprise various types of prosthetic heart valves although the object may comprise other prostheses such as a prosthetic bone fragment, or other prosthetic implant. In further examples, the object may comprise pacemakers or other medical devices or medical implants or other surgical objects to be percutaneously delivered into inserted into the vasculature of the patient.

[0062] In some aspects, the expandable introducer **101** can further comprise an outer tubular layer **401** comprising an inner surface **417** bonded to the inner tubular layer **403**. In some aspects, as shown in FIGS. 3-4, the outer tubular layer **401** can comprise an optional elongated split **301** extending along the radially expandable region **407** in a direction of the elongated axis **201**. As shown in FIG. 4, the optional elongated split **301** can be formed by a first edge facing a second edge. As shown, the first edge can comprise an unobstructed view of the second edge wherein an empty space exists between the first edge and the second edge when the radially expandable region **407** of the inner tubular layer **403** is in the contracted orientation. Although not shown, in some embodiments, the first edge and the second edge of the optional elongated split **301** may abut one another or overlap one another in the contracted orientation in further embodiments. Furthermore, as shown, the optional elongated split **301** can extend along the radially expandable region **407** in the direction of the elongated axis **201** while the radially expandable region **407** of the inner tubular layer **403** is positioned entirely radially inward relative to the optional elongated split **301**.

Although not shown, in further embodiments, the radially expandable region **407** of the inner tubular layer **403** may partially extend within the optional elongated split **301** wherein the radially expandable region **407** partially or entirely obstructs the view of the first edge from the second edge. Still further, the first edge and the second edge are illustrated as parallel with respect to one another although the first edge (or portions of the first edge) may not be parallel to the second edge (or portions of the second edge) in further embodiments. As further illustrated, the first edge and the second edge are each linear edges that are parallel with one another and each extend in the direction of the elongated axis **201**. In further embodiments, although not shown, the first edge and the second edge may comprise curvilinear or other non-linear edge shapes.

[0063] The optional elongated split **301** can be configured such that when an object **3101** passes through the inner lumen **405**, the optional elongated split **301** circumferentially expands to allow the object **3101** to pass therethrough. As shown by joining segment **402** schematically illustrated in FIG. 4, any of the embodiments of the disclosure may provide the outer tubular layer **401** without the previously-described elongated split. Rather, as schematically represented by the joining segment **402**, the outer tubular layer **401** may comprise a cross section taken perpendicular to the elongated axis **201** that is continuous around the entire periphery without any breaks that would otherwise be caused by an elongated split. Furthermore, such continuity can extend along a substantial portion of the length, such as the entire length, of the outer tubular layer **401** in a direction of the elongated axis **201**. Furthermore, while the joining segment **402** is illustrated as having a thickness that is substantially the same as the thickness as the remainder of the periphery of the outer tubular layer, different thicknesses may be provided in further embodiments. For instance, the thickness of the joining segment **402** can be thinner at the radially expandable region **407** to allow easier expansion of the joining segment **402** at the foldable region **409** during expansion of the inner tubular layer **403**.

[0064] The outer tubular layer **401** may be a wide range of materials. For example, the outer tubular layer **401** may be formed from a shape memory material or a resilient material, such as for example, polyurethane (e.g., Pellethane®, Elasthane™, Texin®, or Tecothane®), Pebax®, polyethylene or the like. Additionally, and/or alternatively, the outer tubular layer **401** may further include material(s) that allow the outer tubular layer **401** to be radiopaque to facilitate the detection of the expandable introducer **101** by electromagnetic radiation.

[0065] Referring to FIGS. 3 and 5, in some aspects, a distal segment **1001** of the inner tubular layer

403 extends beyond a distal end **306** of the outer tubular layer **401**. In some aspects, the expandable introducer **101** further comprises a flexible tubular member **300** that comprises a proximal end **305** that can abut, and in some examples engage, the distal end **306** of the outer tubular layer **401** although the proximal end **305** of the flexible tubular member **300** may not abut the distal end **306** of the outer tubular layer **401** (e.g., may be spaced apart) in further embodiments. The flexible tubular member **300** further comprises a distal end **307**, and a length **L1** extending from the proximal end **305** to the distal end **307** in a direction of the elongated axis **201**. For example, the length **L1** can be from about 5 mm to about 7 mm. In aspects, the flexible tubular member **300** can comprise a flexible tip **313** that distally extends beyond a distal end **315** of the inner tubular layer **403**. The flexible tip **313** can comprise a length **L2** extending from the distal end **315** of the inner tubular layer **403** to the distal end **307** of the flexible tubular member **300** in the direction of the elongated axis **201**. For example, the length **L2** of the flexible tip **313** can be from about 0.5 mm to about 3 mm, for example, from about 1 mm to about 1.5 mm.

[0066] The flexible tubular member **300** may be made from a variety of suitable materials. For example, the flexible tubular member **300** can be made from a highly flexible elastic material having a high columnar strength. High columnar strength refers to the ability of the material to withstand buckling or collapsing. In this way, the material should permit the flexible tubular member **300** and the flexible tip **313** to expand and contract, enabling them to accommodate the object **3101** passing therethrough while being capable of preventing the flexible tip **313** from collapsing. For example, a high columnar strength will prevent the flexible tip **313** from collapsing inward towards the proximal end **305** of the flexible tubular member **300** when the object **3101** is proximally retracted relative to the flexible tubular member **300** when removing the object **3101** from the patient's vasculature and from the expandable introducer **101**. The material can be a wide range of materials. For example, the flexible tubular member **300** may be made from Thermoplastic Polyurethane (TPU), Pellethane, Texin, and/or any other similarly suitable materials.

[0067] As shown in FIG. 5, in some aspects the flexible tubular member **300** comprises a bonded region **501** and a flexible region **311** (schematically represented by dashed lines). The bonded region **501** and flexible region **311** are each schematically represented by dashed lines in FIG. 5 that are intended to represent an example of the circumferential extent of the bonded and flexible regions at an arcuate interface between the flexible tubular member **300** and the distal segment **1001** of the inner tubular layer **403**. For example, bonded region **501** is understood to comprise an inner surface **503** of the flexible tubular member **300** bonded to an outer surface **504** of the distal segment **1001** of the inner tubular layer **403** along the circumferential portion of the arcuate interface between the inner surface **503** and the outer surface **504** that is parallel to the illustrated arcuate broken line indicating the bonded region **501** of the arcuate interface. Furthermore, in some embodiments, the bonded region can extend a length (e.g., the entire length) of the distal segment **1001** of the inner tubular layer **403** in the direction of the elongated axis **201**.

[0068] As further shown in FIG. 5, flexible region **311** is understood to comprise the inner surface **503** of the flexible tubular member **300** that is not bonded to the outer surface **504** of the distal segment **1001** of the inner tubular layer **403** along the circumferential portion of the arcuate interface between the inner surface **503** and the outer surface **504** that is parallel to the illustrated arcuate broken line indicating the flexible region **311** of the arcuate interface. As such, within the flexible region, the inner surface **503** of the flexible tubular member **300** is not bonded to the radially expandable region **407** of the inner tubular layer **403**. Furthermore, as shown in FIG. 3, the flexible region **311**, in some embodiments, can extend a length (e.g., the entire length) of the distal segment **1001** of the inner tubular layer **403** in the direction of the elongated axis **201**.

[0069] In aspects, the radially expandable region **407** of the distal segment **1001** of the inner tubular layer **403** extends under the flexible region **311** of the flexible tubular member **300**. For example, the radially expandable region **407** can extend under the flexible region **311** of the flexible tubular member **300** along the entire length of the distal segment **1001** in the direction of

the elongated axis **201**. In this way, when the object **3101** passes through the inner lumen **405**, the flexible region **311** and the radially expandable region **407** will be able to simultaneously radially expand together to accommodate the object **3101** passing therethrough.

[0070] In the embodiment illustrated in FIG. 5, the flexible tubular member **300** surrounds the inner tubular layer **403** with an inner cross-sectional surface profile (see inner surface **503**) matching an outer cross-sectional surface profile (see outer surface **504**) of the distal segment **1001** of the inner tubular layer **403**. In this way, the flexible tubular member **300** can be bonded to the inner tubular layer **403** within the bonded region **501** as discussed above. In some aspects, the flexible tubular member **300** comprises a circular cross-sectional area **505** with a uniform wall thickness **507** circumscribing the distal segment **1001** of the inner tubular layer **403**.

[0071] In some aspects, the bonded region **501** can be formed utilizing a number of suitable bonding processes. For example, the bonding process may include but is not limited to a reflow bonding process or a laser bonding process, such as for example, by pressing the flexible tubular member **300** to the outer cross-sectional surface profile of the distal segment **1001** of the inner tubular layer **403** and applying a heat source to form the bonded region **501**. In this way, the bonded region **501** is formed where the flexible tubular member **300** becomes joined to the outer cross-sectional surface profile of the distal segment **1001** where the heat source is applied. The bonding procedure is not limited to a reflow or laser bonding process. For example, the bonding procedure may comprise an adhesive bonding process or a process that involves manufacturing the flexible tubular member **300** as a monolithic structure of the expandable introducer **101**.

[0072] FIG. 6 illustrates additional embodiments, of a flexible tubular member **600** comprising an inner circumferential flared segment **601** comprising a flared end **603** abutting a folded end of the outer fold **411** of the foldable region **409**. In some embodiments, the inner circumferential flared segment **601** may extend (e.g., continuously extend) the entire length “L1” of the flexible tubular member **600** (see “L1” in FIG. 3). Alternatively, the inner circumferential flared segment **601** may extend less than the entire length of the flexible tubular member **600**. For example, the inner circumferential flared segment **601** may extend along the distal segment **1001** of the inner tubular layer **403** without extending into the length L2 of the flexible tip **313**.

[0073] As further shown in FIG. 6, the flared end **603** can comprise a thickness that gradually circumferentially increases (e.g., in the illustrated counterclockwise direction) toward the outer fold **411**. Further, the outer fold **411** circumferentially extends toward the flared end **603** of the flared segment (e.g., in the illustrated clockwise direction) such that the folded end of the outer fold **411** faces the flared end **603**. As shown, in some embodiments, the folded end of the outer fold **411** can abut the flared end **603** wherein the flared end **603** can provide a seat for the folded end of the outer fold **411**. The inner circumferential flared segment **601** can fill a negative space that may otherwise be formed by the outer fold **411**. In this way, as shown in FIG. 6, the inner circumferential flared segment **601** may provide the outer circumferential surface of the flexible tubular member **600** with a more uniform profile and can support the flexible tubular member **600** to resist radial deformation and stress concentrations that may otherwise impact the profile of the flexible tubular member **600**. The circumferential flared segment **601** can therefore provide a more uniform or circumferential expansion of the flexible tubular member **600**.

[0074] FIG. 7 illustrates an additional embodiment of a flexible tubular member **700** comprising a first inner protrusion **701a** aligned with the folded end of the outer fold **411** of the foldable region **409** and a second inner protrusion **701b** aligned with the folded end of the inner fold **413** of the foldable region **409**. In some embodiments, the first and second inner protrusion **701a**, **701b** may extend (e.g., continuously extend) the entire length “L1” of the flexible tubular member **700** (see “L1” in FIG. 3). Alternatively, the first and second inner protrusion **701a**, **701b** may extend less than the entire length of the flexible tubular member **700**. For example, the first and second inner protrusion **701a**, **701b** may extend along the distal segment **1001** of the inner tubular layer **403** without extending into the length L2 of the flexible tip **313**.

[0075] As further illustrated in FIG. 7, the first and second inner protrusions **701a**, **701b** can comprise thickened areas of the flexible tubular member **700** that are spaced apart from one another by an intermediate area of reduced thickness relative to the first and second inner protrusions **701a**, **701b**. Furthermore, as shown, in some embodiments, a majority of the circumference of the flexible tubular member **700** can extend with a reduced thickness relative to the first and second inner protrusions **701a**, **701b** that circumferentially extends from the first inner protrusion **701a** to the second inner protrusion **701b**. As shown, the majority of the circumference of the flexible tubular member **700** and the intermediate area extending between the first and second inner protrusions **701a**, **701b** can comprise substantially the same reduced thickness although different thicknesses may be provided in further embodiments. As shown, the first inner protrusion **701a** can be aligned to press against the folded end of the outer fold **411** while the second inner protrusion **701b** can be aligned to press against the folded end of the inner fold **413**. In this way, a more uniform circumferential expansion of the flexible tubular member **700** can be maintained subsequent to the bonding process.

[0076] The flexible tubular member **300**, **600**, **700** can be manufactured utilizing any number of manufacturing methods. For example, the flexible tubular member **300**, **600**, **700** can be created utilizing an extrusion process, and/or a molding process. In some embodiments, the flexible tubular member **300**, **600**, **700** can be formed by radio-frequency molding, or radio-frequency forming of an extrusion. In this way, the flexible tubular member **300**, **600**, **700** can be manufactured to conform to a variety of material characteristics and design specifications, such as for example, a material with a high columnar strength and low modulus of elasticity to facilitate the expansion and contraction of the flexible region **311** while avoiding failure (e.g., collapsing).

[0077] As shown in the cross-section of FIG. 8, in some embodiments, the flexible tip **313** can comprise a circular cross-sectional area **800** with a uniform wall thickness **803** although nonuniform wall thicknesses may be provided in further embodiments. In other aspects, the flexible tip **313** may comprise any other shape(s) required to accommodate the object **3101** passing through the distal end **315** of the inner tubular layer **403**.

[0078] FIGS. 9-12 illustrate exemplary methods of manufacturing the expandable introducer **101** in accordance with the present disclosure. FIG. 9 illustrates a perspective view of the inner tubular layer **403** comprising the foldable region **409** comprising the outer fold **411** and the inner fold **413**. Further depicted is the inner lumen **405** extending along the elongated axis **201**. In aspects, the foldable region **409** can extend along the elongated axis **201** of the inner lumen **405** (e.g., along the entire length of the inner tubular layer **403** in the direction of the elongated axis **201**). In some aspects, a gap can be formed between the outer fold **411** and the inner fold **413** and extends along the elongated axis **201**.

[0079] As explained previously, during a reflow or laser bonding process, heat may be applied to surfaces of materials to be bonded. In this way, the layers of materials begin to melt as a result of the directed heat source. Accordingly, the surfaces of the materials begin to contact one another. Once the layers of the melted materials solidify, the layers form a strong bond with one another.

[0080] In aspects, in order to prevent the bonding of the foldable region **409**, for example, portions of the flexible tubular member **300** from melting and flowing into the inner fold **413** and/or portions of the inner fold **413** and the outer fold **411** melting together, an inner shim **910** can be placed within the gap between the outer fold **411** and the inner fold **413**. For example, the inner shim **910** can be slidably positioned through the gap between the outer fold **411** and the inner fold **413** and extend into the inner fold **413** from outside of the inner tubular layer **403**. The inner shim **910** can extend along the elongated axis **201** of the inner lumen **405** any length required to prevent a bonding of the inner fold **413** together and/or a bonding of the outer fold **411** to the inner fold **413**. For example, the inner shim **910** can extend a total length of the inner tubular layer **403**. In an alternative embodiment, the inner shim **910** can extend half the length of the inner tubular layer **403**. In some examples, as shown in FIG. 10, the inner shim **910** can have a length that is greater

than the length of the distal segment **1001** in the direction of the elongated axis **201**. For example, as shown, in some embodiments, the proximal end of the inner shim **910** can extend proximally past the distal end **306** of the outer tubular layer **401** while also extending distally past the distal end **315** of the inner tubular layer **403**.

[0081] FIG. **10** illustrates a perspective view of the expandable introducer **101** comprising the outer tubular layer **401** positioned over the inner tubular layer **403** with the optional elongated split **301** of the outer tubular layer **401** aligned with the foldable region **409** of the inner tubular layer **403**. Although not illustrated in FIG. **10**, in further embodiments, the outer tubular layer **401** may comprise a circular cylindrical tube without an elongated split. Indeed, the outer tubular layer **401** may continuously extend as schematically represented by the joining segment **402** illustrated in FIGS. **4** and **30-31**. Furthermore, as discussed above, the joining segment **402** may have a reduced thickness and can be aligned with the foldable region **409** of the inner tubular layer **403** to facilitate expansion of the joining segment **402** at the location where the inner tubular layer **403** unfolds during expansion of the inner tubular layer **403**. As further illustrated, the outer tubular layer **401** can be positioned relative to the inner tubular layer such that the distal segment **1001** of the inner tubular layer **403** extends beyond the distal end **306** of the outer tubular layer **401**.

[0082] In some aspects, the flexible tubular member **300**, **600**, **700**, shown in FIGS. **2-3**, and **5-8**, can be bonded to the distal segment **1001**. The distal segment **1001** can be any number of suitable lengths. For example, the distal segment **1001** can be a length of 6 mm. In this way, the flexible tubular member **300**, **600**, **700** will have a 6 mm bonding length. In this non-limiting example, the proximal end **305** of the flexible tubular member **300** can abut the distal end **306** of the outer tubular layer **401** and extend distally in the distal direction along the elongated axis **201** of the inner lumen **405** as described in FIGS. **2-3**, and **5**.

[0083] FIG. **11** illustrates a perspective side view of the expandable introducer **101** with an outer shim **1102** comprising a proximal portion positioned on a distal portion of an outer surface **1103** of the outer tubular layer **401** and extending along the elongated axis **201** of the inner lumen **405** and over an outer area of the distal segment **1001**. For example, as shown, in some embodiments, the outer shim **1102** can extend over the outer area along the entire length of the distal segment **1001**. Due to the positioning of the outer shim **1102**, the flexible tubular member **300** will not bond to the distal segment **1001** of the inner tubular layer **403** at the outer area covered by the outer shim **1102** and the flexible tubular member will not bond the facing split edges of the outer tubular layer **401** together. In this way, the flexible region **311** will be created to allow the foldable region **409** to freely unfold and facing edges to spread apart so that the inner tubular layer **403** can achieve the expanded orientation. In some aspects, the bonded region **501** may be calculated to bond to half the circumference of the distal segment **1001** to the flexible tubular member **300**. For example, the circumferential extent of the bonded region **501** can be calculated by determining the circumference of the distal segment **1001** and dividing the determined circumference number in half. In other aspects, the outer shim **1102** can be positioned to cover the folded end of the outer fold **411** and the folded end of the inner fold **413**. In other aspects, the outer shim **1102** can be any other suitable width required to form the flexible region **311**.

[0084] In some aspects, the outer shim **1102** can be a variety of suitable lengths. For example, in aspects, the outer shim **1102** can correspond to the length of the distal segment **1001**, such as for example, in some embodiments the distal segment **1001** can be 6 mm from the distal end **306** of the outer tubular layer **401** and the outer shim **1102** can also be 6 mm and extend distally from the distal end **306** of the outer tubular layer **401**. In other non-limiting examples, the outer shim **1102** can be any other length necessary to form the flexible region **311**. In the illustrated embodiment, the outer shim **1102** can be longer than the length of the distal segment **1001** to facilitate removal of the outer shim **1102** after bonding the flexible tubular member to the distal segment **1001**. For example, as mentioned previously, a proximal portion of the outer shim **1102** can be positioned on the distal portion of the outer surface **1103** of the outer tubular layer **401**. At the same time, in some

embodiments, the outer shim can extend along the elongated axis **201** of the inner lumen **405** while extending over the outer area of the distal segment, and distally past the distal end **315** of the inner tubular layer **403**. The outer shim **1102** can therefore prevent undesired bonding at the foldable region **409** and the distal portion of the elongated split **301**, if provided. As such, once formed, the foldable region **409** is free to unfold while the facing edges of the elongated split **301**, if provided, are free to spread apart so that the inner tubular layer **403** can achieve the expanded orientation. [0085] FIG. **12** illustrates a perspective view of the expandable introducer **101** shown in FIG. **11** with the flexible tubular member **300, 600, 700** bonded to the distal segment **1001** of the inner tubular layer **403**. As shown, the flexible tip **313** of the flexible tubular member **300, 600, 700** extends beyond the distal end **315** of the inner tubular layer **403**. The inner shim **910** and outer shim **1102** have been removed since the bonding process has been completed. For example, after bonding the flexible tubular member **300, 600, 700** to the distal segment **1001**, the inner shim **910** may be gently pulled out from between the gap of the outer fold **411** and the inner fold **413**, utilizing extra precaution to avoid tearing or ripping the outer fold **411** or the inner fold **413**. The inner shim **910** and the outer shim **1102** may be any number of suitable materials that prevent the previously-described bonding. For example, the inner shim **910** and/or the outer shim **1102** can be made from a thin sheet of ceramic, metal, polyimide, PTFE and/or the like.

[0086] In some aspects, FIGS. **13, 15, and 17** show perspective views of example support members **1300, 1500, 1700** that can be provided with the distal segment **1001** to increase the columnar strength of the inner tubular layer **403**, the flexible tubular member **300, 600, 700**, and/or the outer tubular layer **401**. Indeed, providing the inner tubular layer **403** with the support member **1300, 1500, 1700** can inhibit, such as prevent, the flexible tubular member **300, 600, 700**, the inner tubular layer **403**, and/or the outer tubular layer **401** from collapsing inwardly (e.g., buckling), such as when an object **3101** is proximally moved back through the distal end **315** and into the inner lumen **405** to be proximally removed from the proximal end of the expandable introducer **101**. In some examples, it may be especially beneficial to provide the distal segment **1001** with the support member **1300, 1500, 1700** when larger objects **3101** (e.g., objects with a diameter that is larger than the inner diameter of the inner lumen **405** in the contracted orientation) are expected to be inserted through the expandable introducer **101**.

[0087] As shown in FIGS. **13, 15, and 17**, the support member **1300, 1500, 1700** can comprise a wire structure **1302, 1502, 1702**. In some embodiments, the wire structure can comprise an elongated wire that is extruded or drawn to form the elongated wire that extends along an elongated axis of the elongated wire. The wire structure can comprise any number of suitable cross-sectional areas along a cross-section taken perpendicular to the elongated axis of the elongated wire. For example, suitable cross-sectional areas can comprise a circular cross-sectional area, a square cross-sectional area, an octagonal cross-sectional area, or other polygonal or curvilinear cross-sectional areas.

[0088] As further shown in FIGS. **13, 15, and 17**, the wire structure **1302, 1502, 1702** can be arranged in a tubular configuration. As shown, the support member **1300, 1500, 1700** can comprise a circular cylindrical tubular configuration although a frustoconical cylindrical tubular configuration can be provided in further embodiments. Still further, as shown, the wire structure **1302, 1502, 1702** can be provided with a series of peaks and valleys arranged in a circumferential sinusoidal pattern.

[0089] For illustration purposes, FIGS. **14, 16, and 18** illustrate the wire structure **1302, 1502, 1702** of FIGS. **13, 15, and 17** in a two-dimensional planar state. As shown, in some aspects, the wire structure **1302, 1502, 1702** can comprise a continuous wire structure although the wire structure can comprise segments that are joined together in further embodiments. In some embodiments, a wire can be continuously bent or otherwise formed into one of the 2-dimensional shapes illustrated in FIGS. **14, 16, and 18**, and then the bent wire structure can be wound around a cylindrical mandrel into the tubular shape illustrated in FIGS. **13, 15, and 17**. The two free ends of the wire

structure can then be integrally attached together (e.g., by welding) to form the support member **1300, 1500, 1700**. In another embodiment, the wire can be bent around a cylindrical mandrel to form the peaks and valleys while also forming the tubular shape illustrated in FIGS. **13, 15** and **17**. The two free ends of the wire structure can then be integrally attached together (e.g., by welding) to form the support member **1300, 1500, 1700**. In still further embodiments, the support members **1300, 1500, 1700** can be fabricated by 3-dimensional printing, casting, machining, or other forming techniques to obtain to form the support member **1300, 1500, 1700**.

[0090] With additional reference to FIGS. **19-21**, in some aspects, the plurality of peaks **1402** of the wire structure **1302** can be circumferentially distributed around the elongated axis **201**. For example, the support member **1300** can be circumferentially distributed around the entire elongated axis **201** (e.g., distributed 360 degrees around the elongated axis **201**). For example, as shown in FIG. **19**, the support member **1300** is shown circumscribing the elongated axis **201** from a top view of the expandable introducer **101** (e.g., with a view of the flexible region **311**). FIG. **20** shows the support member **1300** circumscribing the elongated axis **201** from a bottom view of the expandable introducer **101**. In this way, the support member **1300** will provide support to both the flexible region **311** and the bonded region **501** (see FIG. **5**). In further aspects, the plurality of peaks **1402** can further comprise a plurality of distal peaks **1404** and a plurality of proximal peaks **1406**.

[0091] In some aspects, each peak of the plurality of distal peaks **1404** can converge in a distal direction (see FIGS. **19** and **20**). For example, an apex of each peak of the plurality of distal peaks **1404** can be pointing towards the distal direction (e.g., in a direction towards the distal end **307** of the flexible tubular member **300, 600, 700**). In some aspects, each peak of the plurality of proximal peaks **1406** can converge in a proximal direction (see FIGS. **19** and **20**). For example, an apex of each peak of the plurality of proximal peaks **1406** can be pointing towards the proximal direction (e.g., in a direction towards the distal end **306** of the outer tubular layer **401**). In some examples, each distal peak of the plurality of distal peaks **1404** can be laterally positioned between a corresponding adjacent pair of proximal peaks of the plurality of proximal peaks **1406**.

[0092] In some aspects, the plurality of distal peaks **1404** and the plurality of proximal peaks **1406** can comprise V-shaped peaks. In some non-limiting examples, with reference to FIG. **14**, any distal peak of the plurality of distal peaks **1404** can comprise a first wire segment **1410** extending from the distal peak in a first direction **1411** to connect to a corresponding first adjacent proximal peak and a second wire segment **1414** extending from the distal peak in a second direction **1415** to connect to a corresponding second adjacent proximal peak. In further non-limiting examples, the above described plurality of proximal peaks **1406** and distal peaks **1404** can be arranged to form the wire structure **1302** in a sinusoidal pattern of alternating distal and proximal peaks circumferentially arranged to circumscribe the elongated axis **201**.

[0093] In some aspects, the plurality of distal peaks **1404** and the plurality of proximal peaks **1406** can comprise rounded peaks, pointed peaks, flat peaks and/or any other suitable peaks. In some examples, one or more distal peaks of the plurality of distal peaks **1404** can comprise different distal peaks than that of the other distal peaks. For example, one distal peak of the plurality of distal peaks can be a V-shaped distal peak (e.g., a pointed peak), while another different distal peak of the plurality of distal peaks **1404** can be a flat distal peak. Any suitable combination of distal peaks (e.g., V-shaped peaks, rounded peaks, pointed peaks, flat peaks) may be utilized. In some examples, one or more proximal peaks of the plurality of proximal peaks **1406** can comprise different proximal peaks than that of the other proximal peaks. For example, one proximal peak of the plurality of proximal peaks **1406** can be a V-shaped proximal peak (e.g., a pointed peak), while another different proximal peak of the plurality of proximal peaks **1406** can be a flat proximal peak. Any suitable combination of proximal peaks for the plurality of proximal peaks **1406** (e.g., V-shaped peaks, rounded peaks, pointed peaks, flat peaks) may be utilized. In some aspects, the plurality of distal peaks **1404** and the plurality of proximal peaks **1406** can comprise the same combination of peaks, such as those described above (e.g., V-shaped peaks, rounded peaks, pointed

peaks, flat peaks). In other aspects, the plurality of distal peaks **1404** and the plurality of proximal peaks **1406** can comprise a different combination of peaks (e.g., V-shaped peaks, rounded peaks, pointed peaks, flat peaks). For example, the plurality of distal peaks **1404** can comprise V-shaped peaks while the plurality of proximal peaks **1406** can comprise flat peaks, or any other suitable combination, such as those described above.

[0094] As shown in FIGS. **19** and **20**, the support member **1300** can extend along the axial length of the distal segment **1001** of the inner tubular layer **403** in a direction of the elongated axis **201**. For example, the support member **1300** can extend along the elongated axis **201** from the distal end **315** of the inner tubular layer **403** to the distal end **306** of the outer tubular layer **401**. In other examples, the support member **1300** can extend along only a portion of the axial length of the distal segment **1001** of the inner tubular layer **403**. For example, the support member **1300** can extend from the distal end **315** of the inner tubular layer **403** to half the axial length of the distal segment **1001**. In other examples, the support member can extend any suitable distance along the axial length of the distal segment **1001** of the inner tubular layer **403** (e.g., more than half the axial length or less than half the axial length).

[0095] In some aspects, as best shown in FIG. **21**, a circumferential distance between each adjacent pair of peaks of the plurality of peaks **1402** can be substantially equal in an unexpanded orientation. For example, as shown in FIG. **21**, a first circumferential distance **2102** between a first distal peak **2106** (represented generally by a dot) and an adjacent second distal peak **2108** can be substantially equal to a second circumferential distance **2104** between the first distal peak **2106** and an adjacent third distal peak **2110** in an unexpanded orientation. In some examples, all adjacent circumferential distances between adjacent distal peaks can be substantially equal in an unexpanded orientation as shown in FIG. **21**. Likewise, all circumferential distances between adjacent proximal peaks can be substantially equal in an unexpanded orientation. Furthermore, in some embodiments, all adjacent circumferential distances between adjacent distal peaks and adjacent proximal peaks can be substantially equal in an unexpanded orientation as will be appreciated by the embodiment illustrated in FIGS. **19-21**.

[0096] FIG. **14** further illustrates the distance between each adjacent pair of peaks of the plurality of peaks **1402** being substantially equal. For example, FIG. **14** shows dotted lines **1418** bisecting the first distal peak **2106**, the second distal peak **2108**, and the third distal peak **2110** of the plurality of distal peaks **1404**. The first circumferential distance **2102** and the second circumferential distance **2104** are shown respectively from the first distal peak **2106** to the second distal peak **2108** and from the first distal peak **2106** to the third distal peak **2110**. While FIG. **14** shows the wire structure **1302** in a two-dimensional planar state, the circumferential distance between adjacent pairs of peaks will also be equal when the wire structure **1302** is in a tubular shape as previously described.

[0097] FIGS. **15-16** illustrate a support member **1500** that can have features, unless otherwise stated, that are similar or identical to features of the support member **1300** illustrated and described with reference to FIGS. **13-14**. As shown in FIGS. **15** and **16**, in some aspects, the wire structure **1502** of the support member **1500** can comprise a plurality of peaks **1602** arranged in a sinusoidal pattern but with a smaller pitch (e.g., the distance from one peak to an adjacent peak) than the sinusoidal pattern of peaks **1402** of the wire structure **1302** of the support member **1300**. In this way, the plurality of distal peaks **1604** and/or the plurality of proximal peaks **1606** can comprise a greater number of peaks per unit length (e.g., total length, half length, and/or the like) of the wire structure **1502** than that of the plurality of distal peaks **1404** and/or the plurality of proximal peaks **1406** per unit length (e.g., total length, half length, and/or the like) of the wire structure **1302**. Increasing the number of peaks per unit length may be useful to increase the rigidity of the flexible tubular member **300**, **600**, **700**, the inner tubular layer **403**, and/or the outer tubular layer **401**. In this way, the flexible tubular member **300**, **600**, **700**, the inner tubular layer **403**, and/or the outer tubular layer **401** will have a greater resistance to buckling, such as for example, while removing an

object **3101** from the expandable introducer **101**.

[0098] FIGS. **17** and **18** illustrate another embodiment of a support member **1700** that can have features, unless otherwise stated, that are similar or identical to features of one or both of the support members **1300**, **1500** illustrated and described with reference to FIGS. **13-16**. For example, the support member **1700** can comprise a wire structure **1702** comprising a plurality of peaks **1802** arranged in a sinusoidal pattern. In some similar aspects to the support members **1300**, **1500**, the plurality of peaks **1802** can be circumferentially distributed around an elongated axis **201** (see FIGS. **22-24**). In further aspects, the plurality of peaks **1802** can further comprise a plurality of distal peaks **1804** and a plurality of proximal peaks **1806**.

[0099] However, as shown in FIGS. **17** and **18**, in some aspects a first portion **1808** (shown generally by the broken-line box in FIG. **18**) of the wire structure **1702** can comprise a smaller pitch (e.g., a distance between peaks) than a second portion **1810** (shown generally by the broken-line box in FIG. **18**) of the wire structure **1702**. In some examples, as shown in FIG. **18**, a pitch P1 between two adjacent distal peaks **1804** of the first portion **1808** can be lesser than a pitch P2 between two different adjacent distal peaks **1804** of the second portion **1810**. As further shown in FIG. **18**, in some embodiments, the pitch between two adjacent proximal peaks **1806** of the first portion **1808** can be the same as the pitch P1 between two adjacent distal peaks **1804** of the first portion **1808**. As further illustrated, in some embodiments, the pitch between two adjacent proximal peaks **1806** of the second portion **1810** can be the same as the pitch P2 between two different adjacent distal peaks **1804** of the second portion **1810**.

[0100] In some aspects, a number of peaks of the plurality of peaks **1802** (e.g., distal peaks **1804** and/or proximal peaks **1806**) in the bonded region **501** can be greater than a number of peaks (e.g., distal peaks **1804** and/or proximal peaks **1806**) of the plurality of peaks **1802** in the flexible region **311**. For example, as shown in FIG. **24**, the flexible region **311** can comprise less distal peaks **1804** per unit of circumferential length than the bonded region **501**. Accordingly, more or less peaks can be provided in accordance with this embodiment described above. Furthermore, it should be understood that while FIG. **24** shows a cross-sectional area of FIG. **22** along a portion of the expandable introducer **101** where the plurality of distal peaks **1804** would be, and thus the description describes the aforementioned principles in terms of distal peaks, the same may be true for the plurality of proximal peaks **1806** (e.g., having less peaks in the flexible region than the bonded region).

[0101] As further shown in FIGS. **22** and **23**, the portion of the support member **1700** that comprises a greater pitch (e.g., pitch P2 of the second portion **1810**) can be positioned around the elongated axis **201** such that the portion of the support member **1700** with less peaks (e.g., the second portion **1810**) can be circumferentially adjacent to the flexible region **311** (see FIG. **22**). In contrast, in some aspects, the portion of the support member **1700** that comprises a smaller pitch (e.g., pitch P1 of the first portion **1808**) can be positioned around the elongated axis **201** such that the portion of the support member **1700** with more peaks (e.g., the first portion **1808**) can be circumferentially adjacent to the bonded region **501** (see FIG. **22**). This may be especially beneficial to provide the flexible tubular member **300**, **600**, **700**, the inner tubular layer **403**, and/or the outer tubular layer **401** with greater rigidity (e.g., to resist buckling when removing an object **3101** from the expandable introducer **101**) in the bonded region where circumferential expansion is not a concern. Furthermore, an increased pitch (decreased number of peaks per unit circumferential length) can be provided in the flexible region to provide reduced circumferential stiffness in the flexible region **311** to avoid an undesired resistance to circumferential expansion in the flexible region **311** while still providing a sufficient level of rigidity to avoid buckling.

[0102] Turning to FIGS. **25-27**, as shown, in some aspects, the support member **1300**, **1500**, **1700** can be limited to only circumferentially extending along a portion (e.g., the bonded region **501**) of the flexible tubular member **300**, **600**, **700**. More specifically, FIG. **26** shows the support member **1300**, **1500**, **1700** circumferentially extending only along the bottom of the expandable introducer

101 (e.g., along a portion of the bonded region **501**). As illustrated in FIG. 25, the support member **1300, 1500, 1700** does not circumferentially extend into the flexible region **311**. In this way, as shown in FIG. 27, the plurality of peaks **1402, 1602, 1802** will only be circumferentially distributed around the elongated axis **201** circumferentially adjacent to the bonded region **501**. It should be understood that while the support member **1300, 1500, 1700** is described above as being tubular, the support member **1300, 1500, 1700** can be adapted to only circumferentially extend along the bonded region **501** in a number of suitable ways. For example, a length (e.g., relative to the two-dimensional planar state shown in FIGS. 14, 16, and 18) or circumference (e.g., relative to the tubular state shown in FIGS. 13, 15, and 17) of the support member **1300, 1500, 1700** can be reduced such that it only extends along the bonded region **501**. In this way, the support member **1300, 1500, 1700** will be an arcuate shape extending along the bonded region **501**. The embodiment of FIGS. 25-27 can provide maximum stiffness in the bonded region that may be sufficient to resist buckling while avoiding any resistance to expansion of the flexible region by the support member **1300, 1500, 1700** to facilitate passage of the object through the inner lumen. [0103] As shown in FIGS. 28 and 29, in aspects, the support member **1300, 1500, 1700** can be biased from radially flaring outward by a constriction of the flexible tubular member **300, 600, 700**. For example, in some aspects, the support member **1300, 1500, 1700** can be configured to self-expand from a constricted state to its natural, expanded state. As shown in FIG. 28, the flexible tubular member **300, 600, 700** can constrict the support member **1300, 1500, 1700** to its contracted state and therefore bias the support member **1300, 1500, 1700** against radially flaring outward. When the flexible tubular member **300, 600, 700** expands (e.g., when an object **3101** enters the flexible tubular member **300, 600, 700**) the support member **1300, 1500, 1700** will tend to self-expand towards its natural, expanded state, as shown in FIG. 29. Thus, support member **1300, 1500, 1700** can be provided, as described above, to increase the rigidity to inhibit, such as prevent, buckling while the tendency of the support member **1300, 1500, 1700** to self expand from the contracted state can help reduce the force necessary for the object to expand the flexible region. [0104] In some aspects, as shown in FIG. 29, the distal end **2802** of the support member **1300, 1500, 1700** can comprise a greater diameter than the proximal end **2804** of the support member **1300, 1500, 1700** when in its natural, expanded state. In some aspects, when the support member **1300, 1500, 1700** is in the constricted state (e.g., as a result of the flexible tubular member **300, 600, 700** constricting the support member **1300, 1500, 1700**), the proximal end **2804** and the distal end **2802** can comprise the same or similar diameters. In some aspects, when the support member **1300, 1500, 1700** is expanded towards its natural, expanded state (e.g., when an object **3101** enters the flexible tubular member **300, 600, 700**), the flexible tubular member **300, 600, 700** can comprise a distally flared frustoconical surface **2902**. In aspects, the distally flared frustoconical surface **2902** extend from the proximal end **305** of the flexible tubular member **300, 600, 700** to the distal end **315** of the inner tubular layer **403**. In this way, for example, the flexible tubular member **300, 600, 700** can gradually increase in diameter from the proximal end **305** of the flexible tubular member **300, 600, 700** to the distal end **315** of the inner tubular layer **403** along the distally flared frustoconical surface **2902**.

[0105] In some aspects, when the support member **1300, 1500, 1700** comprises an arcuate shape (as described above with reference to FIGS. 25-27) to extend only along the bonded region **501**, the proximal end **2804** and the distal end **2802** can comprise same or similar arc lengths when in the constricted state. In some such examples, the distal end **2802** will radially flare out relative to the proximal end **2804** when the support member **1300, 1500, 1700** is expanded towards its natural, expanded state.

[0106] Furthermore, in some such examples, the distally flared frustoconical surface **2902** can circumferentially extend around only a portion of the flexible tubular member **300, 600, 700** (e.g., less than 360 degrees around the flexible tubular member **300, 600, 700**). For example, the distally flared frustoconical surface **2902** can circumferentially extend only around the bonded region **501**

of the flexible tubular member **300, 600, 700**.

[0107] In some aspects, the support member **1300, 1500, 1700** can be heat set to create a self-expanding material. In some aspects, the distal end **2802** and the proximal end **2804** of the support member **1300, 1500, 1700** can be heat set to impart the above described radial flare. For example, by heat setting the material of the support member **1300, 1500, 1700** in a conical shape, the support member **1300, 1500, 1700** will be biased to return to its conical shape. In this way, the support member **1300, 1500, 1700** will exhibit a propensity to naturally revert from the constricted state to its natural, expanded state.

[0108] In some aspects, by making the support member **1300, 1500, 1700** a self-expanding material, it can help to reduce the amount of force required to stretch the flexible region **311** of the flexible tubular member **300, 600, 700** (e.g., when an object **3101** enters the flexible tubular member **300, 600, 700**). For example, because the support member **1300, 1500, 1700** will have a tendency to want to radially expand but is prohibited from doing so by the constriction of the flexible tubular member **300, 600, 700**, the support member **1300, 1500, 1700** can decrease the amount of force required for an object **3101** to stretch the flexible region **311** of flexible tubular member **300, 600, 700** by providing an additional source of radial expansion. Furthermore, as the distal end is flared outwardly, in the natural, expanded state, even less force is necessary at the distal end when proximally retracting the object back through the catheter at the end of the procedure, thereby minimizing the potential of buckling at the point of entry.

[0109] In some aspects, the support member **1300, 1500, 1700** can comprise nitinol. In some aspects, the nitinol can provide a self-expanding support member **1300, 1500, 1700**. This is not meant to be limiting, any other suitable material may be utilized for the support member **1300, 1500, 1700**.

[0110] In some aspects, the support member **1300, 1500, 1700** can be embedded within the flexible tubular member **300, 600, 700**. For example, the flexible tubular member **300, 600, 700** can be formed with the support member **1300, 1500, 1700** contained therein, such as for example, by injection molding the flexible tubular member **300, 600, 700** around the support member **1300, 1500, 1700**. Any other suitable method known in the art may be utilized to embed the support member **1300, 1500, 1700** within the flexible tubular member **300, 600, 700**. In some aspects, the support member **1300, 1500, 1700** can be embedded within the inner tubular layer **403**. For example, the support member **1300, 1500, 1700** can be embedded within the distal segment **1001** of the inner tubular layer **403**. In other aspects, the support member **1300, 1500, 1700** can be provided as an insert and circumferentially positioned within the inner lumen **405**, such as for example, circumferentially positioned within the inner lumen **405** and extending along the distal segment **1001** (e.g., from distal end **315** of the inner tubular layer **403** to the proximal end **305** of the flexible tubular member). In some such examples, the insert can be circumferentially positioned between the flexible tubular member and the distal segment **1001**. For example, the support member **1300, 1500, 1700** can be circumferentially positioned over top of the distal segment **1001** of the inner tubular layer **403** (e.g., not embedded within the distal segment **1001**) and running underneath the flexible tubular member **300, 600, 700**. In some such examples, the flexible tubular member **300, 600, 700** can be bonded (e.g., such as described previously) to the distal segment **1001** of the inner tubular layer **403**, after circumferentially positioning the support member **1300, 1500, 1700** around the distal segment **1001**. In other aspects, the support member **1300, 1500, 1700** can be circumferentially positioned over top and around the flexible tubular member **300, 600, 700**. In yet another aspect, the support member **1300, 1500, 1700** can be embedded within a portion of the flexible tubular member **300, 600, 700**, such as for example, a circumferential upper portion and/or a lower portion of the flexible tubular member **300, 600, 700** (e.g., such as to only be distributed around the bonded region **501** as described with reference to FIGS. 25-27).

[0111] In some aspects, the support member **1300, 1500, 1700** can axially extend past the proximal end **305** of the flexible tubular member **300, 600, 700**. For example, the support member **1300,**

1500, 1700 can axially extend into and/or over a portion of the outer tubular layer **401** (e.g., can be embedded within a portion of the outer tubular layer **401** and/or contact a portion of the outer tubular layer **401**). In further such examples, the support member **1300, 1500, 1700** can be positioned within the inner lumen **405** and/or embedded into the inner tubular layer **403**, as described above, and axially extend past the proximal end **305** of the flexible tubular member **300, 600, 700** and into the outer tubular layer **401**. In yet another such example, the support member **1300, 1500, 1700** can be embedded within or positioned overtop of the flexible tubular member **300, 600, 700** and can axially extend past the proximal end **305** of the flexible tubular member **300, 600, 700** (e.g., can be embedded within a portion of the outer tubular layer **401** and/or contact a portion of the outer tubular layer **401**).

[0112] It should be understood that the example pitches of the peaks illustrated and described with reference to FIGS. **13-29** are merely exemplary. Accordingly support members **1300, 1500, 1700** described herein may comprise any range of pitches, such as for example, peaks with a greater or reduced pitch and/or various pitch patterns. Furthermore, it should be realized that the support member **1300, 1500, 1700** can comprise a greater or lesser number of peaks than that shown in FIGS. **13-29**.

[0113] FIGS. **30-35** will now describe a method of guiding the object **3101** through the expandable introducer **101** comprising the flexible tubular member **300, 600, 700** with the understanding that similar or identical methods may be provided in the other embodiments of the disclosure. More specifically, FIG. **35** illustrates an exemplary flow chart showing example steps for guiding the object **3101** through the expandable introducer **101**.

[0114] Referring to FIG. **35**, during step **3501**, an incision is made at an anatomical location of the patient's vessel. For example, a transfemoral approach that involves making an incision in the groin and operates as a passageway for guiding a catheter-based delivery system to a target site (e.g., a diseased or defective heart valve).

[0115] As shown during step **3503**, the method can further comprise percutaneously introducing a guide wire into the patient's vasculature via the incision made during step **3501**. The guide wire can then be carefully guided through the patient's vasculature to the target site. For example, the guide wire can be guided through the incision in the groin of the patient to the location of the diseased or defective heart valve (e.g., aortic valve). The guide wire can serve as a path for subsequent objects to travel along when inserted into the patient via the incision made during step **3501**.

[0116] During step **3505**, the method can comprise percutaneously inserting the expandable introducer **101** into the incision made during step **3501** and sliding the expandable introducer **101** into the patient's vasculature. For example, the expandable introducer **101** can slide along the guide wire introduced during step **3503**. In some aspects, the expandable introducer **101** can have a dilator introduced within the inner lumen **405** to facilitate the expansion of a passageway within the patient's vessel. As shown, in some embodiments, the flexible tip **313** can comprise a constricted conical shape. In some aspects, the constricted conical shape of the flexible tip **313** axially covers the distal end **315** of the inner tubular layer **403** and thereby provides a beneficial atraumatic feature that helps avoid injury or irritation to the patient's vasculature when inserting the expandable introducer **101**.

[0117] Next, during step **3507** the method can comprise distally advancing the object **3101** through the proximal end of the inlet **105** into the proximal end **309** of the inner lumen **405**. As shown in FIGS. **30-31**, a portion **3001** of the radially expandable region **407** radially expands from a contracted orientation to an expanded orientation while a corresponding portion of the outer tubular layer **401** expands to accommodate the expanded orientation or the radially expandable region of the inner tubular layer **403**. In embodiments with an elongated split **301**, the expansion of the corresponding portion of the outer tubular layer **401** can comprise separating the elongated split **301**, as shown in FIGS. **30-31**, to accommodate the expanded orientation of the inner tubular layer **403**. The object **3101** is schematically represented in cross-section in FIGS. **31** and **33**.

Alternatively, if the outer tubular layer **401** is not provided with an elongated split, a portion of the outer tubular layer **401** (e.g., the joining segment **402**) or the entire circumference of the outer tubular layer **401** may expand to accommodate the expanded orientation of the inner tubular layer **403**.

[0118] As shown in FIG. **31**, in embodiments with an elongated split **301**, the radially expandable region **407** of the inner tubular layer **403** can radially expand while the facing edges of the elongated split **301** of the outer tubular layer **401** circumferentially spread apart such that the inner tubular layer **403** achieves the expanded orientation to accommodate the object **3101**. Indeed, as shown, the facing edges of the elongated split **301** can spread apart from one another such that the width of the split increases to accommodate the expansion of the inner tubular layer **403**. The portion **3001** appears bulged as a result of the object **3101** passing through the inner lumen **405** and thus conforms to the shape of the object **3101** passing therethrough. Alternatively, if the outer tubular layer **401** is provided as a continuous tube (e.g., with the joining segment **402**) without the elongated split, portions of the outer tubular layer **401** or the entire outer circumference of the outer tubular layer **401** can stretch to accommodate the expansion of the inner tubular layer **403**. For example, the circumferential width of the joining segment **402** can increase from the width shown in FIG. **4** to the expanded width shown in FIG. **31** to accommodate expansion of the inner tubular layer **403**. In further embodiments, the entire circumferential width of the outer tubular layer **401** (e.g., not just a portion of the outer tubular layer in the vicinity of the foldable region **409**) can stretch to accommodate the expansion of the inner tubular layer **403**. As illustrated in FIG. **31**, the foldable region **409** has fully unfolded. For example, the outer fold **411** and the inner fold **413** have fully unfolded such that the inner lumen **405** has fully radially expanded to its maximum diameter. In some other aspects, the outer fold **411** and the inner fold **413** can partially radially expand (e.g., wherein the inner lumen **405** only partially radially expands). The foldable region **409** can expand to any range of suitable diameters. For example, the foldable region **409** can have an outer fold **411** and an inner fold **413** configured to accommodate the largest diameter object **3101** expected to pass through the inner lumen **405**. For example, a fold length between the outer fold **411** and the inner fold **413** can be increased (e.g., by forming a larger outer fold **411** and/or inner fold **413**). In this way, the maximum expanded outer diameter of the inner tubular layer **403** will be increased, thereby allowing larger objects to pass through the inner lumen **405**. As further illustrated, the elongated split **301** has separated (or the joining segment **402** has expanded) to accommodate the expansion of the radially expandable region.

[0119] After the object **3101** has been introduced into the expandable introducer **101**, during step **3509** the method can subsequently comprise distally advancing the object **3101** past the distal end **306** of the outer tubular layer **401**. The radially expandable region **407** of the inner tubular layer **403** then radially expands along the axial length of the distal segment **1001** while simultaneously circumferentially stretching the flexible region **311** of the flexible tubular member **300**, **600**, **700**. As such, once the object **3101** is advanced past the distal end **306**, the distal segment **1001** of the inner tubular layer **403** and the flexible tubular member **300**, **600**, **700** simultaneously expand to accommodate the object **3101** passing through the inner lumen **405** of the distal segment **1001** of the inner tubular layer **403**. For example, FIGS. **32-33** show the object **3101** being distally advanced past the distal end **306** of the outer tubular layer **401**, wherein a portion **3201** of the expandable introducer **101** radially expands as a result of the expansion of the radially expandable region **407** and the flexible region **311**.

[0120] In some aspects, the distal segment **1001** of the inner tubular layer **403** can be provided with a support member **1300**, **1500**, **1700**. In further aspects, distally advancing the object **3101** past the distal end **306** of the outer tubular layer **401** can further comprise radially expanding the support member **1300**, **1500**, **1700**. In some examples, the support member **1300**, **1500**, **1700** can radially expand simultaneously with the circumferential stretching of the flexible region **311** of the flexible tubular member **300**, **600**, **700**. In other examples, the support member **1300**, **1500**, **1700** can begin

radially expanding prior to the circumferential stretching of the flexible region **311**, such as for example, where the support member **1300, 1500, 1700** is provided as an insert that extends at least partially into a distal end **306** of the outer tubular layer **401**.

[0121] In some aspects, radially expanding the support member **1300, 1500, 1700** can further comprise radially flaring out a distal end **2802** of the support member **1300, 1500, 1700** relative to a proximal end **2804** of the support member **1300, 1500, 1700**. In further aspects, the support member **1300, 1500, 1700** can comprise a self-expanding support member **1300, 1500, 1700**. In some such aspects, radially flaring out the distal end **2802** can flare the support member **1300, 1500, 1700** towards its natural, expanded orientation.

[0122] During step **3511**, the method can further comprise distally advancing the object **3101** past the distal end **315** of the inner tubular layer **403**. For example, after the object **3101** has been distally advanced past the distal end **315** of the inner tubular layer **403**, the object **3101** will begin entering the vasculature of the patient. In some examples, the object **3101** can move past the distal end **315** of the inner tubular layer **403** via a guide wire. In other examples, the object **3101** can move past the distal end **315** of the inner tubular layer **403** and into the vasculature of the patient without a guide wire.

[0123] Next, during step **3513**, as shown in FIG. **34**, subsequent to an enlarged end portion **3102** of the object **3101** passing through the distal end **315** of the inner tubular layer **403**, the flexible tubular member **300, 600, 700** and the inner tubular layer **403** contract towards a substantially similar original orientation as they were prior to the object **3101** passing through. For example, the resiliency of the material allows for the flexible tubular member **300, 600, 700** and the inner tubular layer **403** to recontract towards their original orientation. In this way, by allowing the flexible tubular member **300, 600, 700** and the inner tubular layer **403** to recontract towards their original orientation, the expandable introducer **101** can accommodate a variety of objects featuring a range of dimensions and structural designs (e.g., a catheter-based delivery system, a dilator and/or the like). For example, the recontraction allows for subsequent entry of a smaller object **3101** following the removal of a larger object **3101** (e.g., where the larger object **3101** has a greater diameter than the smaller object **3101**) while reducing or eliminating blood loss.

[0124] During step **3515**, as further illustrated by FIG. **34**, the method can further comprise distally advancing the object **3101** through the flexible tip **313** and past the distal end **307** of the flexible tubular member **300, 600, 700**. In some aspects, the flexible tip **313** forms a rounded transition **3103** at an interface with an outer circumferential surface **3107** of the object **3101** when advancing the object **3101** past the distal end **307** as well as forming a rounded transition at the distal end **315** of the inner tubular layer **403**. The rounded transitions can provide a more atraumatic feature compared to a transition with sharp edges, thereby avoiding irritation and possible vascular damage from the expandable introducer **101**. In some aspects, the distal end **307** of the flexible tubular member **300, 600, 700** forms a circumferential seal **3105** against the outer circumferential surface **3107** of the object **3101** as the object **3101** passes through the flexible tip **313** and past the distal end **307** of the flexible tubular member **300, 600, 700**. The circumferential seal **3105** in combination with the recontraction of the flexible tubular member **300, 600, 700** and the inner tubular layer **403** (as explained with reference to step **3513**) will be especially beneficial in preventing blood loss when an object **3101** is inserted, removed, repositioned, rotated and/or otherwise adjusted. In some aspects, the flexible tip **313** can be formed in the shape of a hollow frustum in order to form the above-referenced atraumatic transitions (i.e., the rounded transition **3103**, and the transition at the distal end **315** of the inner tubular layer **403**).

[0125] As shown in step **3517**, the method can further comprise positioning the object **3101** within a location of the patient. For example, in some aspects, the object **3101** can be a catheter delivery system, and the catheter delivery system can be inserted through the inner lumen **405** and positioned within a location of the patient's heart, for example, the location of the aortic valve (e.g., the position between the left ventricle and the aorta). Subsequently, the catheter-based delivery

system can be deployed, such as for example, releasing a valve retention member from the catheter-based delivery system to deploy a prosthesis into the patient's heart.

[0126] In other embodiments, the method can further comprise proximally retracting the object **3101** within the inner lumen **405** proximally past the distal end **306** of the outer tubular layer **401** and out of the inner lumen **405** of the distal segment **1001** of the inner tubular layer **403**. Referring to FIG. **34**, for example, the enlarged end portion **3102** of the object **3101** can be proximally retracted relative to the flexible tubular member **300, 600, 700**. The flexible tip **313** can be designed to maintain the circumferential seal **3105** against the outer circumferential surface **3107** to inhibit (e.g., prevent) blood from entering into the inner lumen **405**. The flexible tip **313** can be designed to resist buckling and/or inversion when the enlarged distal end of the object **3101** reaches the distal end **307** of the flexible tubular member **300, 600, 700**. Indeed, the distal tip can be fabricated from a material that has a sufficiently high axial stiffness to avoid buckling and/or inversion. Furthermore, the length “L2” can be kept sufficiently low to increase the axial stiffness as compared to longer lengths. For example, the length “L2” of the flexible tip **313** can be from about 0.5 mm to about 3 mm, for example, from about 1 mm to about 1.5 mm to avoid bucking and/or inversion when the enlarged distal end of the object **3101** is proximally retracted into the inner lumen **405** to cause the distal segment **1001** and the flexible tubular member **300, 600, 700** to radially expand to accommodate the object passing through the distal segment **1001** as shown in FIGS. **32-33**. In further embodiments, as shown and described with respect to FIGS. **13-29** above, the flexible tubular member **300, 600, 700** can be provided with the support member **1300, 1500, 1700** to increase the axial stiffness to further resist buckling and/or inversion when the enlarged distal end of the object **3101** reaches the distal end **307** of the flexible tubular member **300, 600, 700**.

[0127] Then, as shown in FIGS. **30-31**, the object **3101** can be further proximally retracted past the distal end **306** of the outer tubular layer **401**. Due to the resiliency of the flexible tubular member **300, 600, 700**, the distal segment **1001** and the flexible tubular member **300, 600, 700** can then radially retract to the original nonexpanded state as shown in FIG. **30**. As further shown in FIGS. **30-31**, the foldable region **409** can unfold to radially expand the portion **3001** of the inner tubular layer **403** while the facing split edges of the elongated split **301** can circumferentially spread apart to further allow passage of the object **3101**. In embodiments without an elongated split **301**, the joining segment **402** or the entire circumference of the outer tubular layer **401** can stretch to further allow passage of the object **3101**. The object **3101** can be further proximally retracted until the object is removed from the proximal end of the expandable introducer **101**. Due to the resiliency of the outer tubular layer **401**, once the object **3101** has passed, the facing edges of the elongated split **301** can be drawn back together (e.g., to the position shown in FIG. **4**), or the joining segment **402** can resiliently retract, wherein the foldable region **409** regains the folded orientation to radially contract the inner tubular layer **403** as shown in FIG. **4**. In embodiments without the elongated split, the outer tubular layer **401** can radially contract to draw back together, wherein the foldable region **409** regains the folded orientation to radially contract the inner tubular layer **403** shown in FIG. **4**.

[0128] In some aspects, when proximally retracting the object **3101** (as described above) within the inner lumen **405** proximally past the distal end **306** of the outer tubular layer **401** and out of the inner lumen **405** of the distal segment **1001** of the inner tubular layer **403**, the flexible tubular member **300, 600, 700** can contract to contract the radially expandable region of the distal segment **1001** of the inner tubular layer **403** as well as the support member **1300, 1500, 1700**. For example, when the object **3101** is past the distal end **306** of the outer tubular layer, due to the resiliency of the flexible tubular member **300, 600, 700**, the flexible tubular member **300, 600, 700** can contract, thereby constricting the support member **1300, 1500, 1700** away from its natural, expanded orientation towards a constricted orientation (e.g., as shown in FIG. **28**).

[0129] In accordance with the disclosure, non-limiting aspects of the disclosure will now be

described. Various combinations of the aspects can be provided in accordance with the disclosure.

[0130] Aspect 1. An expandable introducer apparatus comprising an inner tubular layer comprising an inner lumen and a radially expandable region extending along an elongated axis of the inner lumen and configured to at least partially radially expand relative to the elongated axis from a contracted orientation to an expanded orientation. An outer tubular layer comprising an inner surface bonded to the inner tubular layer, wherein a distal segment of the inner tubular layer extends beyond a distal end of the outer tubular layer. A flexible tubular member comprising a bonded region and a flexible region, wherein an inner surface of the bonded region is bonded to an outer surface of the distal segment of the inner tubular layer, the flexible region is not bonded to the radially expandable region of the inner tubular layer along an axial length of the distal segment, and the radially expandable region of the inner tubular layer along the axial length of the distal segment extends under the flexible region of the flexible tubular member.

[0131] Aspect 2. The expandable introducer apparatus of Aspect 1, wherein the flexible tubular member comprises a proximal end, a distal end, and a length extending from the proximal end to the distal end of the flexible tubular member in the direction of the elongated axis.

[0132] Aspect 3. The expandable introducer apparatus of Aspect 2, wherein the length is from about 5 mm to about 7 mm.

[0133] Aspect 4. The expandable introducer apparatus of any one of Aspects 2-3, wherein the proximal end of the flexible tubular member abuts the distal end of the outer tubular layer.

[0134] Aspect 5. The expandable introducer apparatus of any one of Aspects 2-4, wherein the flexible tubular member comprises a flexible tip that extends beyond a distal end of the inner tubular layer.

[0135] Aspect 6. The expandable introducer apparatus of Aspect 5, wherein the flexible tip comprises a length extending from the distal end of the inner tubular layer to the distal end of the flexible tubular member.

[0136] Aspect 7. The expandable introducer apparatus of Aspect 6, wherein the length of the flexible tip is from about 0.5 mm to about 3 mm.

[0137] Aspect 8. The expandable introducer apparatus of Aspect 7, wherein the length of the flexible tip is from about 1 mm to about 1.5 mm.

[0138] Aspect 9. The expandable introducer apparatus of any one of Aspects 1-8, wherein the radially expandable region of the inner tubular layer comprises a foldable region configured to be at least partially unfolded from the contracted orientation to the expanded orientation.

[0139] Aspect 10. The expandable introducer apparatus of Aspect 9, wherein the foldable region further comprises an outer fold and an inner fold in the contracted orientation.

[0140] Aspect 11. The expandable introducer apparatus of Aspect 10, wherein the flexible tubular member comprises an inner cross-sectional surface profile that matches an outer cross-sectional surface profile of the distal segment of the inner tubular layer.

[0141] Aspect 12. The expandable introducer apparatus of any one of Aspects 10-11, wherein the flexible tubular member comprises an inner circumferential flared segment comprising a flared end abutting a folded end of the outer fold of the foldable region.

[0142] Aspect 13. The expandable introducer apparatus of Aspect 10, wherein the flexible tubular member comprises a first inner protrusion aligned with a folded end of the outer fold of the foldable region and a second inner protrusion aligned with a folded end of the inner fold of the foldable region.

[0143] Aspect 14. The expandable introducer apparatus of any one of Aspects 1-10, wherein the flexible tubular member further comprises a circular cross-sectional area with a uniform wall thickness circumscribing the distal segment of the inner tubular layer.

[0144] Aspect 15. The expandable introducer apparatus of any one of Aspects 1-14, wherein the distal segment of the inner tubular layer is provided with a support member.

[0145] Aspect 16. The expandable introducer apparatus of Aspect 15, wherein the support member

comprises a wire structure.

[0146] Aspect 17. The expandable introducer apparatus of Aspect 16, wherein the wire structure comprises a sinusoidal wire structure comprising a plurality of peaks circumferentially distributed around the elongated axis.

[0147] Aspect 18. The expandable introducer apparatus of Aspect 17, wherein the plurality of peaks further comprise a plurality of distal peaks and a plurality of proximal peaks.

[0148] Aspect 19. The expandable introducer apparatus of Aspect 18, wherein each peak of the plurality of distal peaks converges in a distal direction.

[0149] Aspect 20. The expandable introducer apparatus of Aspect 18, wherein each peak of the plurality of proximal peaks converges in a proximal direction.

[0150] Aspect 21. The expandable introducer apparatus of any one of Aspects 18-20, wherein a distal peak of the plurality of distal peaks comprises a first wire segment extending from the distal peak in a first direction to connect to a corresponding proximal peak of the plurality of proximal peaks, and wherein the distal peak of the plurality of distal peaks comprises a second wire segment extending from the distal peak in a second direction to connect to another corresponding proximal peak of the plurality of proximal peaks.

[0151] Aspect 22. The expandable introducer apparatus of Aspect 21, wherein the first direction comprises a first proximal direction and the second direction comprises a second proximal direction.

[0152] Aspect 23. The expandable introducer apparatus of any one of Aspects 15-22, wherein the support member extends along the axial length of the distal segment of the inner tubular layer in the direction of the elongated axis.

[0153] Aspect 24. The expandable introducer apparatus of any one of Aspects 17-23, wherein a circumferential distance between each adjacent pair of peaks of the plurality of peaks is substantially equal.

[0154] Aspect 25. The expandable introducer apparatus of any one of Aspects 15-24, wherein the support member only circumferentially extends along the bonded region of the flexible tubular member.

[0155] Aspect 26. The expandable introducer apparatus of any one of Aspects 17-24, wherein a number of peaks of the plurality of peaks in the bonded region is greater than a number of peaks of the plurality of peaks in the flexible region.

[0156] Aspect 27. The expandable introducer apparatus of any one of Aspects 15-26, wherein the support member comprises nitinol.

[0157] Aspect 28. The expandable introducer apparatus of any one of Aspects 15-27, wherein the support member is embedded within the flexible tubular member.

[0158] Aspect 29. The expandable introducer apparatus of any one of Aspects 15-28, wherein the support member is biased from radially flaring outward by a constriction of the flexible tubular member.

[0159] Aspect 30. The expandable introducer apparatus of any one of Aspects 1-29, wherein the outer tubular layer comprises an elongated split extending along the radially expandable region in a direction of the elongated axis.

[0160] Aspect 31. A method of guiding an object through an inner lumen of an expandable introducer apparatus, the expandable introducer apparatus comprising: an inner tubular layer comprising the inner lumen and a radially expandable region extending along an elongated axis of the inner lumen; an outer tubular layer comprising an inner surface bonded to the inner tubular layer, wherein a distal segment of the inner tubular layer extends beyond a distal end of the outer tubular layer; and a flexible tubular member comprising a bonded region and a flexible region, wherein an inner surface of the bonded region is bonded to an outer surface of the distal segment of the inner tubular layer, the flexible region is not bonded to the radially expandable region of the inner tubular layer along an axial length of the distal segment, and the radially expandable region

of the inner tubular layer along the axial length of the distal segment extends under the flexible region of the flexible tubular member, the method comprising distally advancing the object through a proximal end of the inner lumen, wherein a portion of the radially expandable region radially expands from a contracted orientation to an expanded orientation while a corresponding portion of the outer tubular layer expands to accommodate the expanded orientation of the radially expandable region. The method further comprising distally advancing the object past the distal end of the outer tubular layer to radially expand the radially expandable region of the inner tubular layer along the axial length of the distal segment while simultaneously circumferentially stretching the flexible region of the flexible tubular member, wherein the distal segment of the inner tubular layer and the flexible tubular member simultaneously expand to accommodate the object passing through the distal segment of the inner tubular layer. Distally advancing the object past a distal end of the inner tubular layer.

[0161] Aspect 32. The method of Aspect 31, wherein a distal end of the flexible tubular member extends distally beyond the distal end of the inner tubular layer to form a flexible tip, and the method further comprises distally advancing the object through the flexible tip and past the distal end of the flexible tubular member.

[0162] Aspect 33. The method of Aspect 32, wherein the flexible tip forms a rounded transition at the distal end of the inner tubular layer when advancing the object past the distal end of the inner tubular layer and through the flexible tip.

[0163] Aspect 34. The method of any one of Aspects 32-33, wherein the distal end of the flexible tubular member forms a circumferential seal against an outer circumferential surface of the object as the object passes through the flexible tip and past the distal end of the flexible tubular member.

[0164] Aspect 35. The method of any one of Aspects 32-34, wherein a length of the flexible tip from the distal end of the inner tubular layer to the distal end of the flexible tubular member is from about 0.5 mm to about 3 mm.

[0165] Aspect 36. The method of any one of Aspects 31-35, further comprising proximally retracting the object within the inner lumen proximally past the distal end of the outer tubular layer and out of the inner lumen of the distal segment of the inner tubular layer, wherein the flexible tubular member contracts to contract the radially expandable region of the distal segment of the inner tubular layer.

[0166] Aspect 37. The method of Aspect 36, wherein the radially expandable region of the distal segment of the inner tubular layer comprises a foldable region, wherein radially contracting the flexible tubular member at least partially folds the foldable region.

[0167] Aspect 38. The method of any one of Aspects 31-36, wherein the radially expandable region comprises a foldable region, wherein radially expanding the radially expandable region at least partially unfolds the foldable region.

[0168] Aspect 39. The method of any one of Aspects 31-38, wherein the distal segment of the inner tubular layer is provided with a support member.

[0169] Aspect 40. The method of Aspect 39, wherein distally advancing the object past the distal end of the outer tubular layer further comprises radially expanding the support member.

[0170] Aspect 41. The method of Aspect 40, wherein the support member radially expands simultaneously with the circumferential stretching of the flexible region of the flexible tubular member.

[0171] Aspect 42. The method of any one of Aspects 40-41, wherein radially expanding the support member further comprises radially flaring out a distal end of the support member relative to a proximal end of the support member.

[0172] Aspect 43. The method of Aspect 42, wherein the support member comprises a self-expanding support member and radially flaring out the distal end flares the support member towards its natural, expanded orientation.

[0173] Aspect 44. The method of any one of Aspects 40-43, further comprising proximally

retracting the object within the inner lumen proximally past the distal end of the outer tubular layer and out of the inner lumen of the distal segment of the inner tubular layer, wherein the flexible tubular member contracts to contract the radially expandable region of the distal segment of the inner tubular layer and the support member.

[0174] Aspect 45. The method of any one of Aspects 31-44, wherein the outer tubular layer comprises an elongated split extending along the radially expandable region in a direction of the elongated axis, and the distally advancing the object through the proximal end of the inner lumen radially expands the portion of the radially expandable region while a corresponding portion of the elongated split separates.

[0175] It should be understood that while various aspects have been described in detail relative to certain illustrative and specific examples thereof, the present disclosure should not be considered limited to such, as numerous modifications and combinations of the disclosed features are possible without departing from the scope of the following claims.

Claims

1. An expandable introducer apparatus comprising: an inner tubular layer comprising an inner lumen and a radially expandable region extending along an elongated axis of the inner lumen and configured to at least partially radially expand relative to the elongated axis from a contracted orientation to an expanded orientation; an outer tubular layer comprising an inner surface bonded to the inner tubular layer, wherein a distal segment of the inner tubular layer extends beyond a distal end of the outer tubular layer; and a flexible tubular member comprising a bonded region and a flexible region, wherein an inner surface of the bonded region is bonded to an outer surface of the distal segment of the inner tubular layer, the flexible region is not bonded to the radially expandable region of the inner tubular layer along an axial length of the distal segment, and the radially expandable region of the inner tubular layer along the axial length of the distal segment extends under the flexible region of the flexible tubular member.
2. The expandable introducer apparatus of claim 1, wherein a proximal end of the flexible tubular member abuts a distal end of the outer tubular layer.
3. The expandable introducer apparatus of claim 1, wherein the flexible tubular member comprises a flexible tip that extends beyond a distal end of the inner tubular layer.
4. The expandable introducer apparatus of claim 3, wherein a length of the flexible tip is from about 0.5 mm to about 3 mm.
5. The expandable introducer apparatus of claim 1, wherein the radially expandable region of the inner tubular layer comprises a foldable region configured to be at least partially unfold from the contracted orientation to the expanded orientation.
6. The expandable introducer apparatus of claim 5, wherein the foldable region further comprises an outer fold and an inner fold in the contracted orientation.
7. The expandable introducer apparatus of claim 1, wherein the distal segment of the inner tubular layer is provided with a support member.
8. The expandable introducer apparatus of claim 7, wherein the support member comprises a wire structure.
9. The expandable introducer apparatus of claim 8, wherein the wire structure comprises a sinusoidal wire structure comprising a plurality of peaks circumferentially distributed around the elongated axis.
10. A method of guiding an object through an inner lumen of an expandable introducer apparatus, the expandable introducer apparatus comprising: an inner tubular layer comprising the inner lumen and a radially expandable region extending along an elongated axis of the inner lumen; an outer tubular layer comprising an inner surface bonded to the inner tubular layer, wherein a distal segment of the inner tubular layer extends beyond a distal end of the outer tubular layer; and a

flexible tubular member comprising a bonded region and a flexible region, wherein an inner surface of the bonded region is bonded to an outer surface of the distal segment of the inner tubular layer, the flexible region is not bonded to the radially expandable region of the inner tubular layer along an axial length of the distal segment, and the radially expandable region of the inner tubular layer along the axial length of the distal segment extends under the flexible region of the flexible tubular member, the method comprising: distally advancing the object through a proximal end of the inner lumen, wherein a portion of the radially expandable region radially expands from a contracted orientation to an expanded orientation while a corresponding portion of the outer tubular layer expands to accommodate the expanded orientation of the radially expandable region; distally advancing the object past the distal end of the outer tubular layer to radially expand the radially expandable region of the inner tubular layer along the axial length of the distal segment while simultaneously circumferentially stretching the flexible region of the flexible tubular member, wherein the distal segment of the inner tubular layer and the flexible tubular member simultaneously expand to accommodate the object passing through the distal segment of the inner tubular layer; and distally advancing the object past a distal end of the inner tubular layer.

11. The method of claim 10, wherein a distal end of the flexible tubular member extends distally beyond the distal end of the inner tubular layer to form a flexible tip, and the method further comprises distally advancing the object through the flexible tip and past the distal end of the flexible tubular member.

12. The method of claim 11, wherein the flexible tip forms a rounded transition at the distal end of the inner tubular layer when advancing the object past the distal end of the inner tubular layer and through the flexible tip.

13. The method of claim 11, wherein the distal end of the flexible tubular member forms a circumferential seal against an outer circumferential surface of the object as the object passes through the flexible tip and past the distal end of the flexible tubular member.

14. The method of claim 11, wherein a length of the flexible tip from the distal end of the inner tubular layer to the distal end of the flexible tubular member is from about 0.5 mm to about 3 mm.

15. The method of claim 10, wherein the radially expandable region comprises a foldable region, wherein radially expanding the radially expandable region at least partially unfolds the foldable region.

16. The method of claim 10, wherein the distal segment of the inner tubular layer is provided with a support member.

17. The method of claim 16, wherein distally advancing the object past the distal end of the outer tubular layer further comprises radially expanding the support member.

18. The method of claim 17, wherein the support member radially expands simultaneously with the circumferential stretching of the flexible region of the flexible tubular member.

19. The method of claim 17, wherein radially expanding the support member further comprises radially flaring out a distal end of the support member relative to a proximal end of the support member.

20. The method of claim 19, wherein the support member comprises a self-expanding support member and radially flaring out the distal end flares the support member towards its natural, expanded orientation.
